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# Urban heat island monitoring and intelligent prediction method based on big data analysis

Lin Wang<sup>1\*</sup>

## Abstract

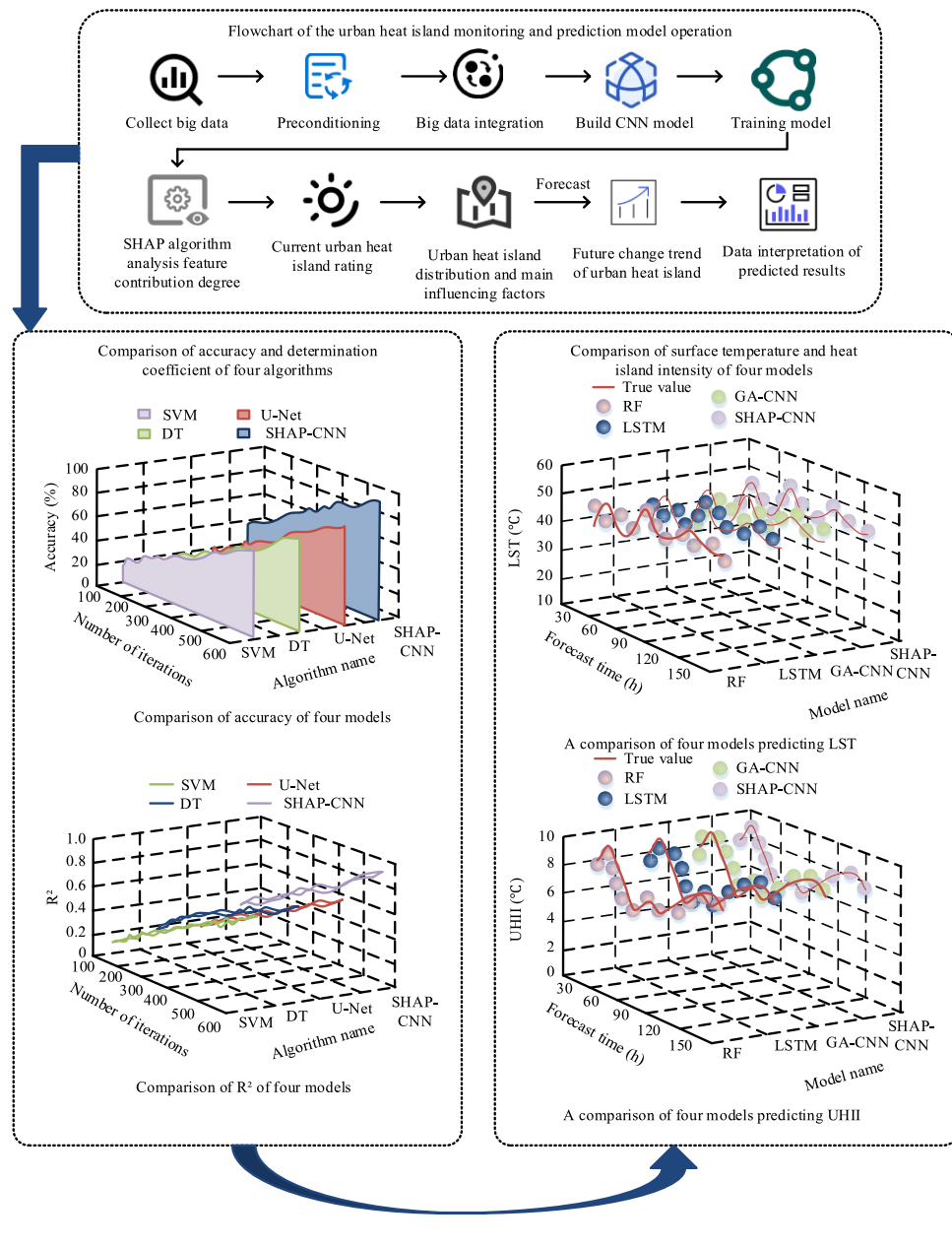
Frequent extreme heat events in urban areas have brought growing public attention to the urban heat island effect. However, traditional monitoring techniques are unable to handle high-dimensional data, which limits their application in real-world scenarios. This paper introduces SHapley Additive exPlanations to optimize Convolutional Neural Networks. The method uses min–max normalization to process big data and applies Gradient Descent to improve the model. It also analyzes the contribution of each heat island factor to the prediction results and generates interpretable evaluation outputs. A data set of urban heat island effect indices is used to validate the model. The results show that the model achieves a final harmonic mean of 0.93 and an accuracy of 88.7%. On a self-built data set of Xiamen's heat island effect, the model reaches a Root Mean Square Error of 0.43 °C, a mean prediction time of 12.3 s, and a median of 11.8 s. The prediction match rate of heat island intensity reaches 99.5%. These results confirm that the model is both reliable and efficient for monitoring and predicting the urban heat island effect. It can make accurate predictions across different urban areas and offers valuable insights for prevention strategies and supports the development of smart city.

**Keywords** Big data, CNN, SHAP, Urban heat island effect, Monitoring model

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## Graphical Abstract



## Introduction

As global urbanization has rapidly advanced, the urban heat island phenomenon has increasingly intensified. Extreme weather threatens public health and seriously affects the quality of urban life (Sachindra et al., 2023). Accurate monitoring and prediction of urban heat island areas have become key research topics (Mishra & Arya, 2024). Traditional approaches often rely on a combination of classical machine learning and statistical analysis.

However, these methods struggle with complex and high-dimensional data, leading to limitations in both efficiency and accuracy (Hong et al., 2022). As artificial intelligence technology rapidly advances, deep learning models offer new solutions for monitoring and predicting the urban heat island effect (Mandal et al., 2022). Some researchers use backpropagation neural networks to analyze the phenomenon and generate predictions through forward propagation. However, these models require

large data sets for training, which makes the process time-consuming and costly, limiting widespread application (Pai et al., 2023). To improve training efficiency, Convolutional Neural Networks (CNN) have become a preferred choice. CNN can automatically extract image features and handle large-scale complex data, especially in the analysis of remote sensing images. In addition, SHapley Additive exPlanations (SHAP) can interpret prediction results and evaluate the contribution of each feature, improving model interpretability. Based on this, the study introduces SHAP to optimize CNN and builds an SHAP–CNN-based model for urban heat island monitoring and prediction. The model uses CNN to automatically extract heat island features and combines SHAP to analyze the influence of big data on the heat island effect. This approach makes the prediction results more interpretable. The aim is to achieve accurate analysis of urban heat islands from a broader perspective, provide more efficient and precise technical support for environmental monitoring, and offer theoretical guidance for policy-making.

The novelty of the research is as follows: first, the combination of SHAP explanatory framework and CNN feature extraction capability solves the "black box" problem of deep learning models in urban climate research; and developed a hybrid fusion algorithm for multi-source heterogeneous data. The second is to establish a new paradigm for predicting urban heat islands through spatiotemporal coupling. Using a gradient optimization-based dynamic convolution kernel adjustment method, the model can simultaneously capture spatial heterogeneity and temporal dynamics. The third is to design a decision oriented analysis framework that can effectively improve the efficiency of practical applications. The fourth is to embed a socio-economic sensitivity analysis module in the model to achieve spatial coupling diagnosis of heat island risk and social vulnerability.

### Related works

The frequent occurrence of elevated temperatures in urban regions has underscored the necessity of precise analysis of the urban heat island effect. This is key to reducing the negative impact of extreme weather on public health. Researchers worldwide integrated intelligent technologies with big data analytics to investigate the urban heat island effect and proposed a variety of solutions. For example, Badugu et al. developed a multi-parameter surface temperature prediction model using long–short-term memory (LSTM) to examine surface temperature variations in India. They explored the influence of surface temperature on the development of the urban heat island effect. The results showed that their model achieved the highest prediction accuracy

and provided a scientific basis for relieving the heat island effect (Badugu et al., 2024). Chen et al. aimed to improve the accuracy of urban thermal environment monitoring. They developed a high-resolution evaluation model of heat island intensity based on Random Forest, using remote sensing images to extract environmental parameters. The experimental findings indicated a correlation coefficient of 0.73 between the predicted and actual temperatures, with a Root Mean Square Error (RMSE) of 0.72 °C, surpassing the performance of the baseline model. (Chen et al., 2022). Barthwal and Sharma designed a mobile sensing system based on the Internet of Things to detect the urban heat island effect. The system monitored multiple regions using location awareness and mobile sensors. Experiments showed that the model achieved the lowest mean absolute error (Barthwal & Sharma, 2022). Badugu et al. estimated surface temperature using thermal remote sensors to study the urban heat island effect in India during summer and winter. The results revealed that the effect was more obvious at night and provided a quantitative basis for understanding the spatial distribution of urban heat islands (Badugu et al., 2023). By employing normalization methods, Cetin et al. analyzed the Normalized Difference Vegetation Index and the Normalized Difference Built-up Index to investigate how surface temperature is related to the urban heat island effect (Cetin et al., 2024). Tanoori et al. used a convolutional neural network model to study the impact of urban structure on land surface temperature (LST) mitigation and land use/cover change. The results show that the formation of different urban heat island patterns is significantly influenced by urban geometry and land use and land cover types. To address the issue of average surface temperature, urban landscape planning should prioritize cohesion, density, and continuity while minimizing fragmentation, variability, and complexity (Tanoori et al., 2024). Li et al. found that recent studies have mainly examined the impact of surface composition and landscape pattern on urban heat islands, but have overlooked examining the connectivity of thermal landscapes from a holistic perspective. Therefore, the study chose to investigate the spatial and temporal development of thermal landscape connectivity at the regional scale, and predicted the spatial pattern of the heat island in 2030 using artificial neural networks, cellular automata, and Markov models. The results showed that the research method simulated a KAPPA coefficient of 0.78, which has high accuracy and can simulate the distribution of thermal landscapes at different time periods (Li et al., 2024). Raza et al. focused on analyzing the heat island phenomenon caused by urban expansion and vegetation changes in three key areas of Punjab Province over the past 30 years (1990–2020): Sheikh Pur, Chiniot,

and Pakpatan. The study adopts a cellular automaton model and combines multiple spatial variables to predict land use and land cover in 2050. The experimental results show that in the past 30 years, the surface temperature of Chiniot has increased by 1.65 °C, Sheikh Pulla has increased by 0.86 °C, and Pakpatan has slightly decreased to -0.04 °C, indicating a negative correlation between LST and vegetation cover (Raza et al., 2024). Zhang et al. designed a new method based on the continuous transition of thermal environment along the urban–rural gradient, which used multiple satellite data sources to quantify the intensity and footprint of surface heat islands in 283 cities in China from 2005 to 2018. The results show that the two-step method avoids the limitations of subjective reference selection and provides reliable quantification of urban SUHI features over time (Zhang & Xia, 2024). Wilson et al. utilized surface temperature (LST), local climate zone (LCZ), as well as deep learning methods such as convolutional neural networks (CNN) and pix2pix to quantify the urban heat island (SUHI) effect on the surface. They extracted and classified the SUHI effect using convolutional fuzzy Q-Gaussian neural networks and variational K–C mean regression learning. The research results indicate that socio-economic variables account for 12% to 20% of the variation in urban heat island intensity (Prakash et al., 2025).

In addition, CNN are widely used in daily life due to their ability to process network-structured data. For example, Hasan et al. applied CNN to solve image classification problems. They trained the model on gray-scale digit images from a handwritten digit data set. The experimental results showed that the CNN model reached a classification accuracy of 98%, confirming its efficiency and precision in image classification tasks (Hasan et al., 2024). Wei et al. applied quantum CNN to process noisy images. They combined basic gates with variational parameters to complete the full image and digit processing task. Experiments confirmed that the model was effective in recognizing noisy images and achieved better accuracy than comparison models (Wei et al., 2022). Sarvamangala et al. introduced CNN to analyze medical images. They refined image classification, segmentation, localization, and detection processes, and applied the model to diagnose brain, breast, and lung diseases. Results showed that CNN could effectively detect medical images and complete classification tasks (Sarvamangala & Kulkarni, 2022). Shah et al. combined artificial neural networks with CNN to build an automated skin cancer detection model. This model used deep learning technology for early diagnosis of skin cancer. Tests on different data sets showed that the hybrid model improved detection accuracy (Shah et al., 2023). Hassan and Maji developed a deep

learning model with initial layers and residual connections to improve the detection of plant diseases. They used depthwise separable convolution to reduce parameter count. The model achieved test accuracies of 99.39%, 99.66%, and 76.59% on three plant disease data sets, performing better than existing models (Hassan & Maji, 2022).

In summary, many existing studies on the urban heat island effect still face problems such as limited prediction accuracy caused by multi-factor interaction and poor model transferability across different cities. Therefore, by integrating the SHAP interpretability framework with deep convolutional networks, research has made the following contributions in the field of urban heat island research: first, a heat island prediction model that supports feature level attribution is proposed, which quantifies the dynamic contributions of factors such as building density and vegetation index to provide causal basis for precise governance. Second, develop a multi-source heterogeneous data fusion architecture to integrate remote sensing images, meteorological sensors, and social media data. Finally, constructing a "monitoring attribution simulation" technology chain to achieve closed-loop analysis from heat island identification to policy evaluation provides a new paradigm for smart city climate adaptation that combines academic rigor and policy practicality.

## Monitoring and prediction model of urban heat island effect based on big data

### Optimized design of the SHAP–CNN fusion algorithm

CNN is an important neural network structure in deep learning. It extracts local spatial features effectively and generates different feature maps through a multi-layer network. Its advantage lies in processing high-dimensional remote sensing images and automatically identifying key spatial information. Therefore, CNN plays an important role in urban heat island monitoring. By analyzing spatial distribution data from satellite remote sensing, CNN can accurately predict urban heat island areas. The core structure of CNN is the convolutional layer, which analyzes the input image through convolution operations, as shown in the following equation:

$$Z = Q(\delta * x + b) \quad (1)$$

In Eq. (1),  $Z$  is the output,  $\delta$  is the weight,  $*$  represents the convolution operation,  $b$  is the bias,  $Q$  is the activation function, and  $x$  is the input data. Since the features come from multiple sources, the data needs to be normalized. Common normalization methods include min–max normalization (min–max norm) and Z-score normalization. This study uses min–max norm to standardize data units, as shown in the following equation:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (2)$$

In Eq. (2),  $X'$  is the normalized data,  $X$  is the original data,  $X_{\min}$  is the minimum value in the data set, and  $X_{\max}$  is the maximum value. The processed data are then fed into CNN for feature analysis. The CNN model updates its parameters by repeatedly calculating the loss function to minimize the error between the predicted and true values, as shown in the following equation:

$$G(\delta) = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{y}_i) \quad (3)$$

In Eq. (3),  $G(\delta)$  is the loss function,  $N$  is the number of samples,  $y_i$  is the true value,  $\hat{y}_i$  is the predicted value, and  $L$  is the loss for each sample. To improve the convergence of CNN, this study uses Gradient Descent (GD) to optimize CNN. Gradient Descent evaluates the loss function's gradient relative to the model parameters and iteratively updates them in the negative gradient direction to achieve loss minimization. The process of GD is shown in Fig. 1.

As shown in Fig. 1, GD first initializes the weights and biases of CNN, and defines the loss function and optimization method. Then, the training data are fed into CNN, and different network structures process the data to obtain predictions. The loss is first computed, followed by backpropagation to determine the gradients, which are then used to update the parameters and minimize the loss. This process repeats until the model finds the optimal parameters. The detailed process is shown in the following equation:

$$\delta = \delta - \alpha \cdot \nabla_{\delta} G(\delta) \quad (4)$$

In Eq. (4),  $\alpha$  is the learning rate, and  $G(\delta)$  is the gradient of the loss function with respect to  $\delta$ . When GD

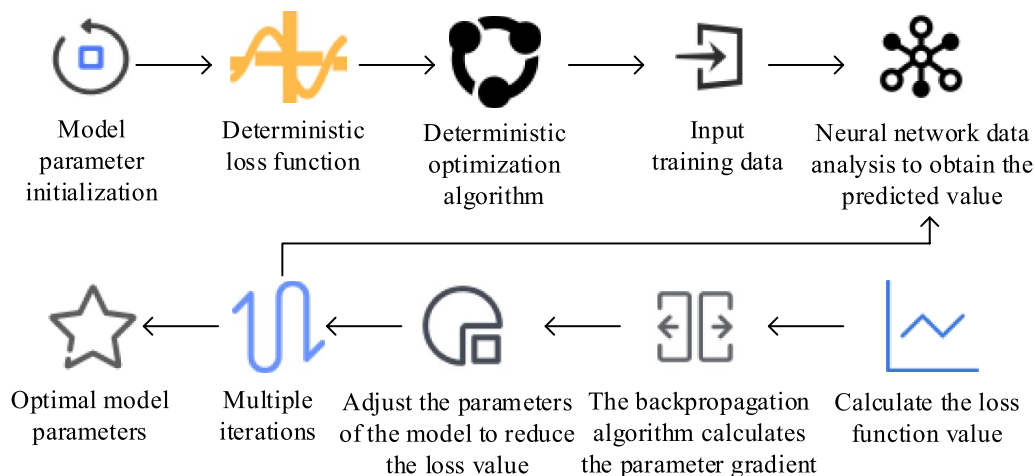
optimizes the weights of CNN, the bias is also updated using the same rule, as shown in the following equation:

$$b = b - \alpha \cdot \nabla_b G(b) \quad (5)$$

In Eq. (5), the model goes through multiple iterations during training. It updates all network parameters through forward and backward propagation to minimize the loss. However, CNN's predictions cannot be traced back to specific input features, making the results less interpretable. SHAP is a game theory-based method for interpreting model predictions. It quantifies the importance of each feature to the prediction results by calculating the marginal contribution of each feature to the model prediction. The SHAP value represents the average impact of a certain feature on the model's prediction results after considering all possible feature combinations. This method can provide global and local explanations, helping researchers understand the decision-making process of the model and identify the most influential features on the predicted results. Therefore, SHAP is introduced to optimize CNN. The SHAP process is shown in Fig. 2.

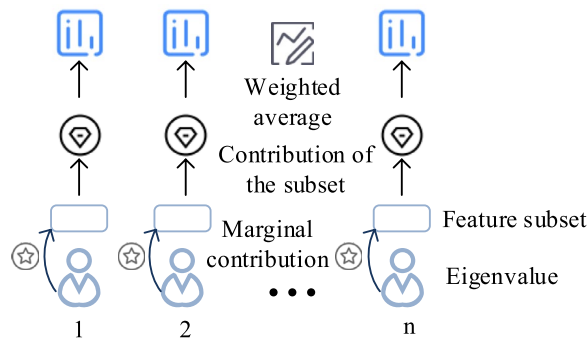
As shown in Fig. 2, SHAP first determines the number of features and constructs feature subsets by considering all possible combinations. It then calculates the marginal contribution of each feature in the subset to determine the subset's impact on the prediction. To ensure fairness, SHAP calculates the weighted average of all subsets to obtain the final contribution of each feature. The SHAP calculation is shown in the following equation:

$$\Phi_i(P) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [P(S \cup \{i\}) - P(S)] \quad (6)$$



**Fig. 1** GD optimization process for CNN





**Fig. 2** Schematic diagram of the SHAP algorithm operation process

In Eq. (6),  $\Phi_i(P)$  is the contribution of feature  $i$ ,  $S$  is the feature subset,  $P(S)$  is the prediction with the subset,  $P(S \cup \{i\})$  is the prediction with the subset plus feature  $i$ ,  $n$  is the number of features, and  $|S|$  is the size of the subset. SHAP is introduced to optimize CNN models with the main objective of improving their interpretability. Although CNN models perform well in processing high-dimensional data and automatically extracting features, their prediction results are difficult to trace back to specific input features, resulting in the "black box" characteristics of the model. By combining SHAP with CNN, the contribution of each input feature to the prediction of urban heat island effect can be quantified, thereby enhancing the interpretability and credibility of the model. After clarifying the SHAP process, the SHAP–CNN hybrid algorithm is constructed, as shown in Fig. 3.

In Fig. 3, the first step is collecting multi-source information of feature objects. The raw data are pre-processed to remove missing values, errors, and noise. Then, all data are normalized using min–max norm and classified based on timestamps. Next, the CNN structure is built and trained on selected data sets. GD is used to update model parameters. Finally, SHAP calculates the contribution of each feature to the prediction, improving the model's interpretability.

### Construction of the SHAP–CNN model for urban heat island monitoring and prediction under big data fusion

This study introduces SHAP to optimize CNN, applies min–max norm for data standardization, and uses GD to enhance CNN training. The SHAP–CNN hybrid algorithm is suitable for analyzing big data and evaluating the importance of different features. It is applied to build a model for real-time urban heat island monitoring and prediction. The model first collects various urban heat island-related data and uses wireless sensor networks to obtain land surface temperature, as shown in the following equation:

$$B_i(t_i) = \varepsilon_i(\theta) \times [\tau_i \times B_i(t_s) + (1 - \tau_i) \times L_i \downarrow] + L_i \uparrow \quad (7)$$

In Eq. (7),  $t_i$  is the temperature of channel  $i$ ,  $t_s$  is the ground temperature,  $\tau_i$  is the ground emissivity,  $\varepsilon_i(\theta)$  is the atmospheric transmittance of channel  $i$  at angle  $\theta$ ,  $B_i(t_i)$  is the received thermal radiation intensity,  $B_i(t_s)$  is the blackbody radiation at ground temperature  $t_s$ ,  $L_i \downarrow$  is the downward atmospheric radiation, and  $L_i \uparrow$  is the upward atmospheric radiation. The blackbody radiation is calculated by Planck's law, as shown in the following equation:

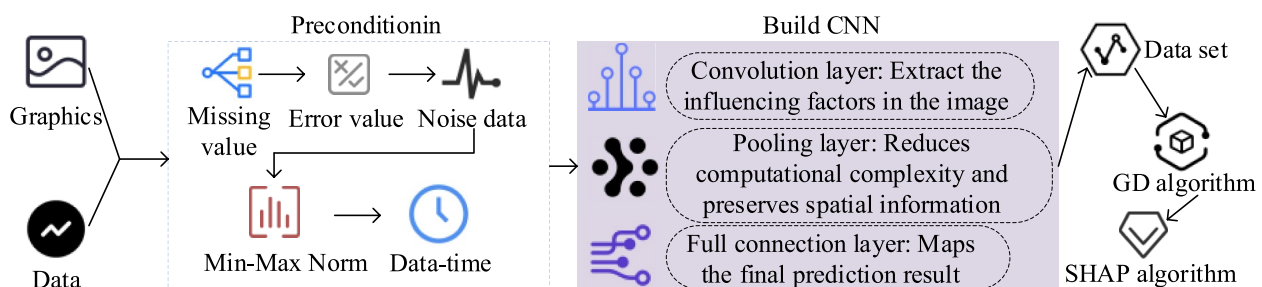
$$B_i(t_s) = [B_i(t_i) - L_i \uparrow - \varepsilon_i(\theta)(1 - \tau_i) \times L_i \downarrow] / \varepsilon_i(\theta) \tau_i \quad (8)$$

Equation (8) gives the blackbody radiation value from Planck's law. Its inverse form is used to calculate the land surface temperature, as shown in the following equation:

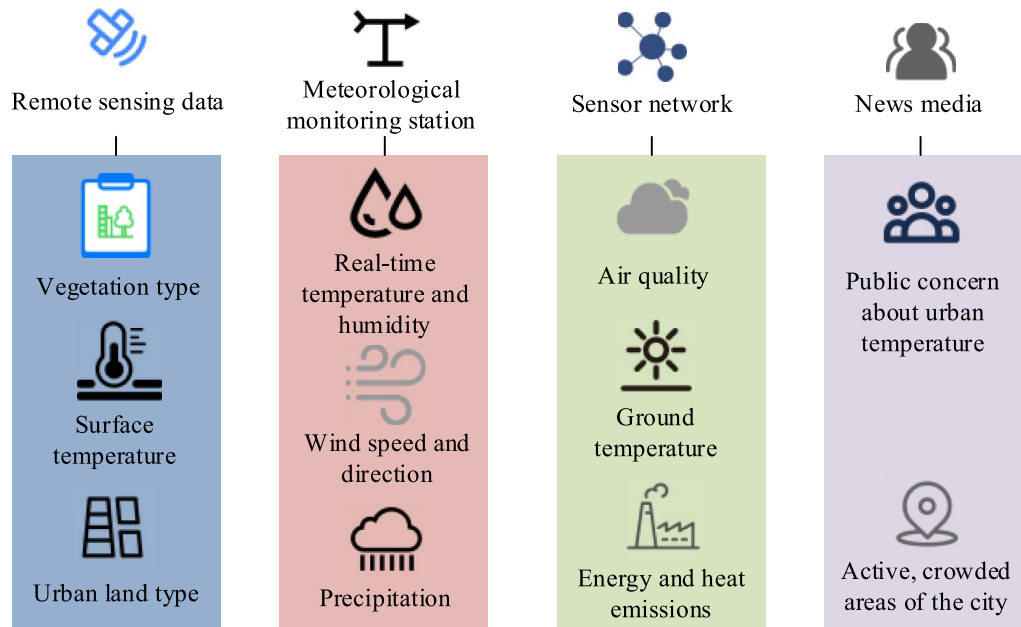
$$T_s = K_2 / \ln[K_1 / B_i(T_s) + 1] - 273.15 \quad (9)$$

In Eq. (9),  $K_1$  and  $K_2$  are constants. After clarifying how surface temperature is obtained, the study collects data on urban heat island-related factors. The data sources are shown in Fig. 4.

As shown in Fig. 4, the data come from four sources: remote sensing, meteorological data, sensor networks, and news media. Remote sensing provides greening types, surface temperatures, and land types. Sensor networks collect air quality data across time to assess daily



**Fig. 3** Flowchart of the SHAP–CNN fusion algorithm



**Fig. 4** Big data sources for building monitoring and prediction models

energy use and heat emissions. News media help understand public concern and identify crowded or active city areas. After identifying these data sources, the fusion method is shown in the following equation:

$$X_f = [X_1, X_2, \dots, X_m] \quad (10)$$

In Eq. (10),  $X_f$  is the fused data, and  $m$  is the number of data items. Because of the diversity of sources, the study uses three fusion methods. The preprocessing and fusion process is shown in Fig. 5.

As shown in Fig. 5, preprocessing focuses on spatial, temporal, and unit consistency. First, spatial coordinates are adjusted. Second, data are classified by time sequence. Third, the data are normalized. After preprocessing, data fusion begins. Feature fusion extracts each feature and merges feature maps. These are then fed into deep networks for further learning. Image fusion stacks land use, temperature, and remote sensing images into multi-channel inputs. Spatiotemporal fusion merges sensor, meteorological, and remote sensing images of the same time period into one image. The study then analyzes urban heat island intensity using the Temperature Difference Index (TDI), as shown in the following equation:

$$T_d = T_c - T_z \quad (11)$$

In Eq. (11),  $T_d$  is the urban heat island intensity,  $T_c$  is the average surface temperature of the city, and  $T_z$  is the average surface temperature of surrounding areas. To further evaluate intensity in functional zones, the calculation is shown in the following equation:

$$Td(C_i) = \frac{A_{h_i}}{A_{C_i}} \times \frac{A_h}{A} \quad (12)$$

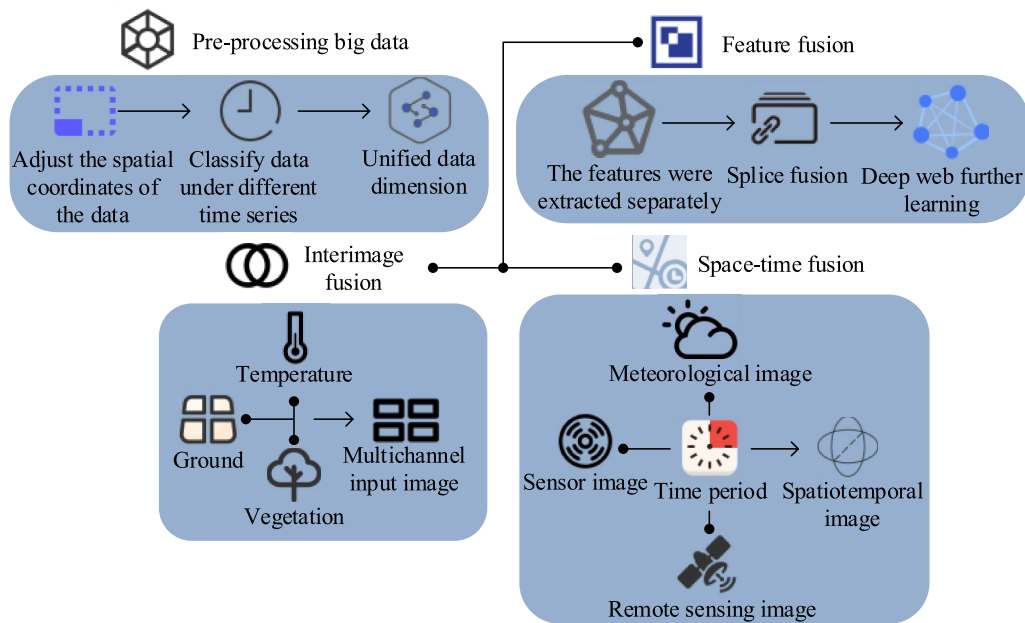
In Eq. (12),  $Td(C_i)$  is the proportion of high-temperature pixels in zone  $i$ ,  $A_{h_i}$  is the pixel area with high surface temperature in zone  $i$ ,  $A_{C_i}$  is the total area of zone  $i$ ,  $A_h$  is the total area of high-temperature pixels in all zones, and  $A$  is the total area of all zones. After clarifying the data fusion and heat island intensity calculation, the model for monitoring and prediction is built. The process is shown in Fig. 6.

As shown in Fig. 6, the model first collects and processes urban heat island data, performs fusion, builds the CNN model, and optimizes the parameters. SHAP then calculates feature contributions and outputs the urban heat island level. A visualization tool is used to show the distribution map and key influencing factors. The model also predicts future changes and explains the results with data support.

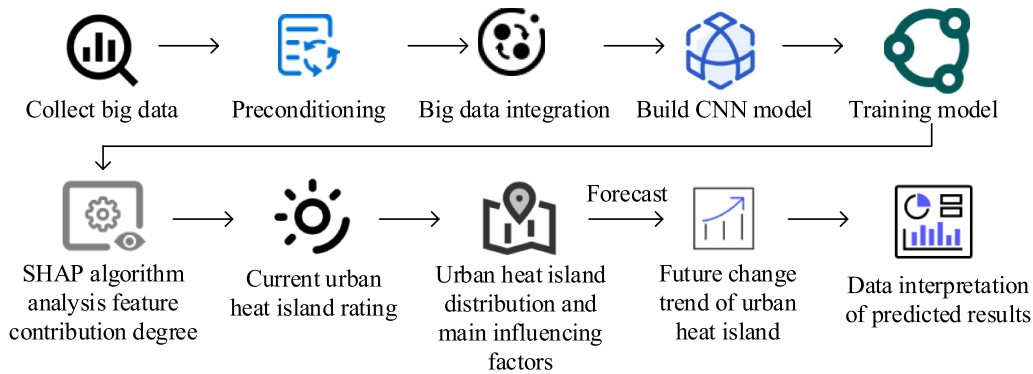
## Performance analysis of the heat island monitoring and prediction model

### Evaluation indicators

To comprehensively and objectively evaluate the performance of urban heat island monitoring and prediction models, multiple performance indicators suitable for classification and regression tasks were selected, including accuracy, F1 value, and recall rate; The regression performance indicators include root mean square variance (RMSE) and coefficient of determination  $R^2$ ,



**Fig. 5** Big data preprocessing and fusion process



**Fig. 6** Flowchart of the urban heat island monitoring and prediction model operation

while the time efficiency indicators include training time and word prediction time; The spatial consistency index includes the matching rate of heat island intensity. The classification index is used to evaluate the ability to distinguish the level of urban heat island; Regression indicators are used to quantify the accuracy of temperature prediction; the time indicator is to ensure the feasibility of actual deployment; the spatial matching rate meets the GIS application requirements of urban planning. The accuracy can intuitively reflect the overall correctness of the model, which is suitable for the task of classifying the intensity level of urban heat islands; The recall rate is used to ensure that high-risk heat island areas are not missed and to safeguard public health; The F1 value can still provide a single and

balanced performance measure when there is an imbalance between positive and negative samples. The specific calculation is shown in the following equation:

$$\begin{cases} Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \\ Recall = \frac{TP}{TP + FN} \\ F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \end{cases} \quad (13)$$

In Eq. (13),  $TP$  represents the number of samples that are actually positive and correctly predicted as positive by the model;  $FP$  represents the number of samples that are actually negative but incorrectly predicted as positive by the model;  $FN$  represents the number of samples that



are actually negative and correctly predicted as negative by the model;  $TN$  represents the number of samples that are actually positive but incorrectly predicted as negative by the model. RMSE is more sensitive to larger errors and directly reflects the magnitude of temperature prediction deviation;  $R^2$  is dimensionless, which facilitates the comparison of model generalization ability across cities and time periods. The expression is shown in the following equation:

$$\begin{cases} RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \\ R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \end{cases} \quad (14)$$

In Eq. (14),  $n$  represents the total number of samples,  $y$  and  $\hat{y}_i$  are the  $i$  true and predicted values of the sample, and  $\bar{y}$  is the average of the true values.

#### Performance comparison of the SHAP–CNN hybrid algorithm

To verify the superiority of the proposed SHAP–CNN hybrid algorithm, this study selected Support Vector Machine (SVM), Decision Tree (DT), and Convolutional Networks for Biomedical Image Segmentation (U-Net) as comparative models. SVM is widely used in the classification and regression tasks of remote sensing images and station meteorological data, especially in small sample scenarios, where it performs robustly; DT has a simple structure and strong interpretability, and is commonly used for rapid baseline comparison and feature importance screening; U-Net is a factual standard in the field of remote sensing semantic segmentation and has been widely used for urban thermal environment mapping. The publicly available urban heat island intensity data set was used as the training data set for the model. This data set contained surface temperature, meteorological data, urban structures, and other related information from multiple cities. Sixty percent of the data set was used as

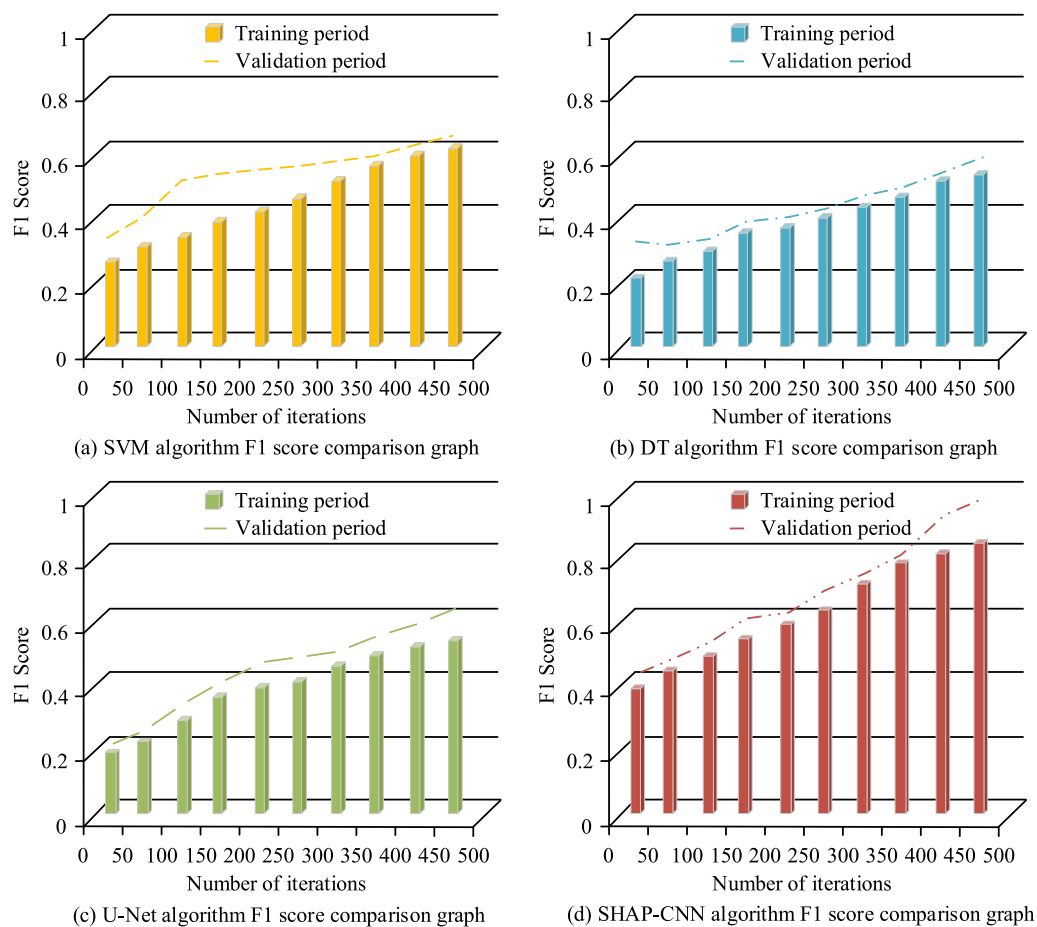
the testing set, 30% as the validation set, and 10% as the result set. A performance testing platform was built for the model, where Windows 11 was used as the operating system, the CPU was Intel Xeon Gold 6248R, the GPU was NVIDIA GTX 1080Ti, and the software environment included Python 3.8 and TensorFlow 2.5. Before conducting the experiment, it is necessary to present the core statistical features of the data set, as shown in Table 1.

According to Table 1, the overall missing rate of the data set is controlled at a relatively low level of 0.8–4.1%. The statistical results show that the average surface temperature is  $32.4 \pm 3.8$  °C, and the building density is  $48 \pm 23\%$ . Both show significant spatial heterogeneity, with a right skewed distribution, indicating the existence of obvious high-temperature abnormal areas in the city; The average vegetation index is  $0.36 \pm 0.12$ , and the left skewed feature reflects the general situation of insufficient urban vegetation coverage; The average intensity of the heat island is  $2.7 \pm 1.1$  °C, and its prominent rightward deviation and peak intensity characteristics reveal the concentrated strengthening phenomenon of the heat island effect in local areas. These statistical features not only verify the spatial differentiation law of urban thermal environment, but also provide important basis for feature engineering in subsequent model construction, especially for the skewed characteristics of data distribution and extreme value aggregation phenomenon, which require appropriate number transformation and spatial attention mechanism to optimize model performance. Meanwhile, the high variability of building density and surface temperature, insufficient vegetation coverage, and spatial clustering characteristics of urban heat island intensity all provide quantitative support for the precise formulation of urban heat island mitigation strategies. The study first selected the F1 score as the evaluation metric and conducted cross validation using the testing and validation sets. The comparison of F1 scores is shown in Fig. 7.

As shown in Fig. 7a, during 100 to 250 iterations, the F1 score in the validation phase was significantly higher than that in the training phase, while the gap between them narrowed in later iterations. In Fig. 7b, the F1 score of the DT algorithm in the training phase started at 0.19,

**Table 1** Core statistical features of the data set

| Statistical indicators | Surface temperature (°C) | Vegetation Index (NDVI) | Building density (%) | Heat island intensity (°C) |
|------------------------|--------------------------|-------------------------|----------------------|----------------------------|
| Sample size            | 28,750                   | 28,750                  | 28,750               | 9520                       |
| Mean $\pm$ SD          | $32.4 \pm 3.8$           | $0.36 \pm 0.12$         | $0.48 \pm 0.23$      | $2.7 \pm 1.1$              |
| Skewness               | 0.82                     | 0.45                    | 0.63                 | 1.05                       |
| Kurtosis               | 2.91                     | 3.12                    | 2.78                 | 3.84                       |
| Missing rate/%         | 1.2                      | 0.8                     | 2.3                  | 4.1                        |



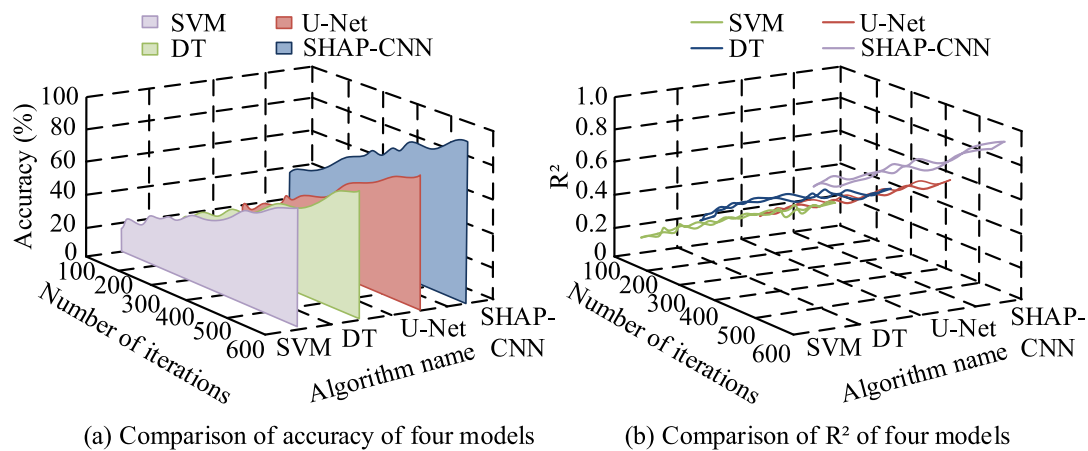
**Fig. 7** F1 score comparison of four algorithms

while it reached 0.29 in the validation phase; however, the model showed limited convergence in later iterations. In Fig. 7c, the U-Net algorithm produced relatively low F1 scores in the early iterations, and after 600 iterations, the F1 score only increased to 0.52. In Fig. 7d, SHAP-CNN achieved an F1 score of 0.81 in the training phase and reached 0.93 in the validation phase. Overall, SHAP-CNN demonstrated a faster improvement in F1 score and outperformed the other algorithms. To conduct a more thorough evaluation of SHAP-CNN's performance, the study analyzed the model based on accuracy and the Coefficient of Determination ( $R^2$ ). The comparison of accuracy and  $R^2$  is shown in Fig. 8.

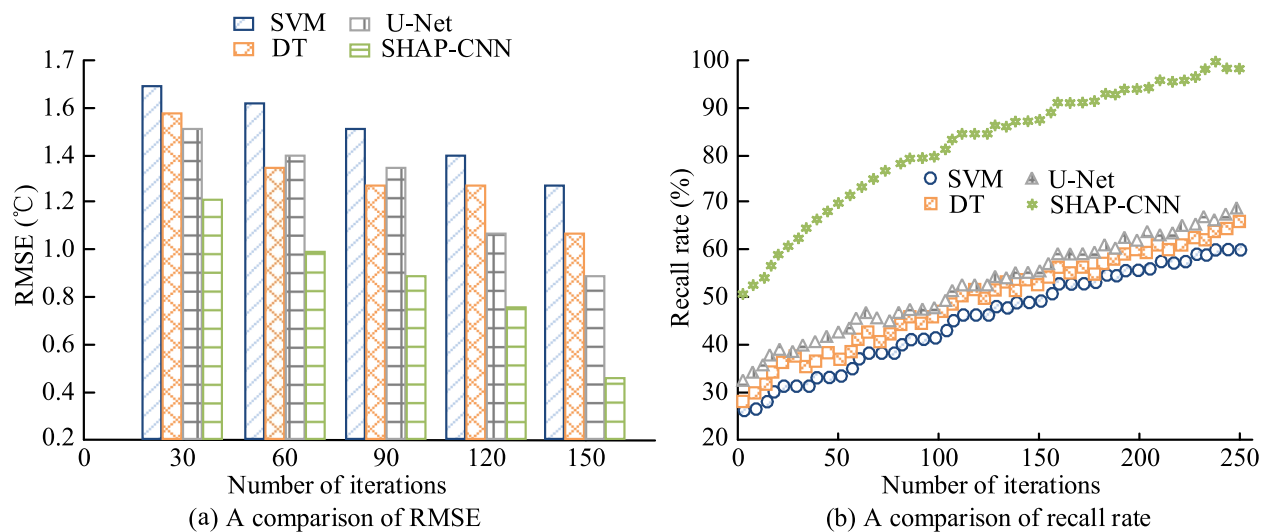
As shown in Fig. 8a, after 600 iterations, the accuracy of the SVM algorithm increased from 18.5% to 53.4%. The U-Net algorithm achieved an accuracy of 62.1% by the end of the iterations. The SHAP-CNN model reached an accuracy of 88.7% after 600 iterations. In Fig. 8b, the  $R^2$  values of SVM and U-Net were relatively low at 100 iterations, reaching 0.11 and 0.15, respectively. SHAP-CNN experienced a temporary low

in  $R^2$  at 400 iterations, but the value continued to rise afterward and stabilized at 0.91 by 600 iterations. These results showed that SHAP-CNN significantly improved the model's prediction accuracy after prolonged training. After confirming the high prediction accuracy of SHAP-CNN, the study introduced RMSE and recall as additional evaluation metrics. The comparison of RMSE and recall for the four algorithms is shown in Fig. 9.

As shown in Fig. 9a, at the end of the iterations, the RMSE of the DT algorithm reached 1.26 °C, which was higher than the 0.89 °C RMSE of the U-Net algorithm. SHAP-CNN showed the most significant drop in RMSE from 1.20 °C to 0.98 °C during iterations 30–60 and eventually stabilized at 0.43 °C. In Fig. 9b, the recall values of SVM, DT, and SHAP-CNN fluctuated between 25.5% and 69.9%. The SVM algorithm achieved the highest recall at the end of the iterations, reaching 69.9%, followed by DT with a recall of 75.1%. SHAP-CNN showed a slower increase in recall during iterations 80–250, rising from 79.2% to 98.8%. Overall, SHAP-CNN demonstrated



**Fig. 8** Comparison of accuracy and determination coefficient of four algorithms



**Fig. 9** Comparison of RMSE and recall rate of four algorithms

higher convergence precision and stability compared to the other algorithms.

#### Performance verification of the SHAP–CNN-based heat island model

After verifying the performance advantages of the SHAP–CNN hybrid algorithm, the study further evaluated the real-world application capability of the SHAP–CNN-based urban heat island monitoring and prediction model. Urban heat island data from Shanghai, a first-tier city in China, and Xiamen, a second-tier city, were collected for evaluation. Fifty percent of the data were used for testing, 30% for validation, and 20% for results. Random forest (RF), LSTM, and genetic algorithm–convolutional neural network hybrid model (GA–CNN) were

selected as comparative models. RF was chosen, because it has robustness to high-dimensional remote sensing features and is easy to interpret; LSTM is adept at capturing long-term dependencies in time series and has been extensively used in predicting the evolution of surface temperature (LST) over time; GA–CNN exhibits high-temperature prediction accuracy in complex urban morphology modeling. Windows 10 was used as the operating system, Intel Xeon Gold 6248R as the CPU, and PyTorch 1.8 as the deep learning framework. Matplotlib and Seaborn were used for visualization. The study first selected RMSE and  $R^2$  of the predicted temperature as comparison metrics and conducted evaluation using data sets from Shanghai and Xiamen. DT and Back Propagation Neural Network (BPNN) were also included in the

comparison. The results for the six models are shown in Table 2.

As shown in Table 2, the SHAP–CNN model achieved the best performance on both Shanghai and Xiamen data sets. On the Shanghai data set, SHAP–CNN yielded an RMSE of 0.48 °C and an  $R^2$  of 0.97. Compared with the GA–CNN model, which had an RMSE of 0.84 °C, the SHAP–CNN model reduced the error by 42.9%. On the Xiamen data set, SHAP–CNN reached an RMSE of 0.43 °C and an  $R^2$  of 0.98. These results showed that the proposed model had the strongest capability in urban heat island prediction and could efficiently complete high-precision monitoring and forecasting tasks. To further evaluate the performance of SHAP–CNN, the study compared the training and prediction time of the models. The comparison is shown in Fig. 10.

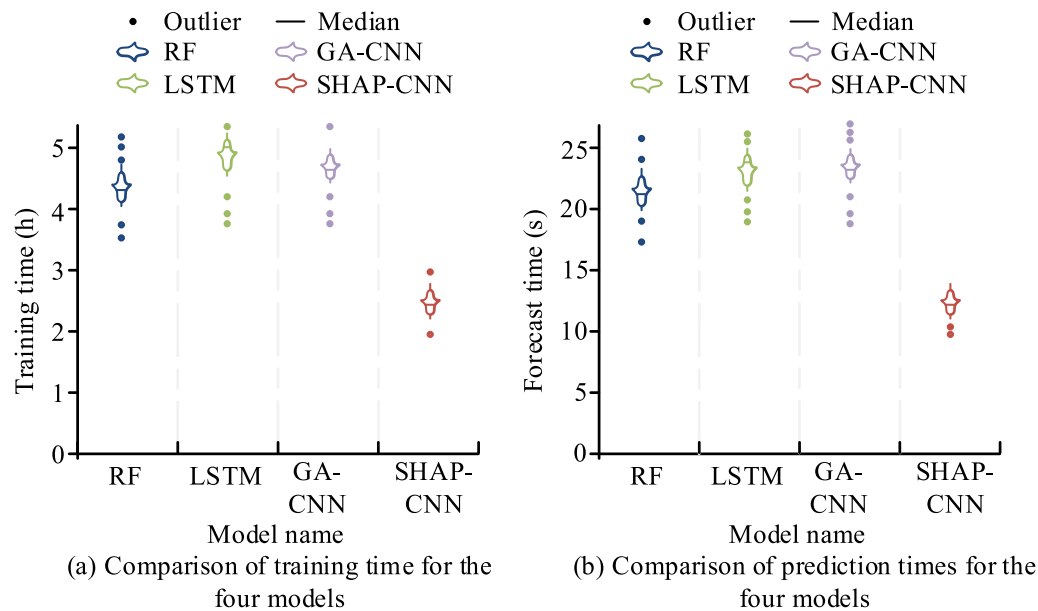
As shown in Fig. 10a, the median training time of RF was 4.28 h, with an average of 4.37 h, which was shorter than that of the LSTM and GA–CNN models. The

median training time of SHAP–CNN was 2.41 h, and the average was 2.49 h. In Fig. 10b, GA–CNN had a median prediction time of 23.1 s, which was shorter than that of LSTM. SHAP–CNN had an average prediction time of 12.3 s, and a median of 11.8 s. Overall, SHAP–CNN achieved shorter training and prediction times than the comparative models, offering high prediction accuracy along with efficient processing. After confirming the convergence and efficiency of the proposed model, the study further analyzed its performance in monitoring and predicting the heat island effect. Surface temperature and heat island intensity were selected as the relevant practical metrics. The comparison results of these two indicators among the four models are shown in Fig. 11.

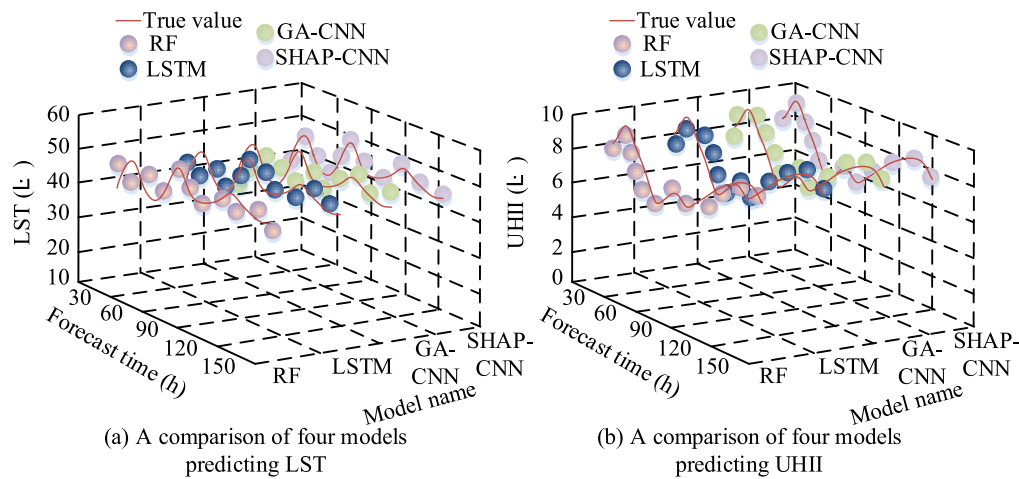
As shown in Fig. 11a, SHAP–CNN had the highest consistency between predicted and actual surface temperature, reaching 98.1%. The LSTM model followed with a matching rate of 82.1%. In Fig. 11b, SHAP–CNN also had the highest accuracy in predicting heat island intensity, achieving a match rate of 99.5%. In contrast, the RF model only reached 70.2%, and both LSTM and GA–CNN tended to produce predicted values that were higher than the actual values. The above results may be due to the fact that SHAP values can help researchers intuitively understand how the model makes predictions based on various features, and observe the direction and degree of influence of a certain feature on the model's prediction results under different values. In the monitoring and prediction of urban heat island effect, it is not only necessary to have accurate prediction results, but also to understand the decision-making basis of the

**Table 2** Comparison of RMSE and  $R^2$  of six models

| Model name | Shanghai  |       | Xiamen |       |
|------------|-----------|-------|--------|-------|
|            | RMSE (°C) | $R^2$ | RMSE   | $R^2$ |
| RF         | 1.11      | 0.72  | 1.08   | 0.86  |
| LSTM       | 1.05      | 0.82  | 0.96   | 0.97  |
| GA–CNN     | 0.84      | 0.85  | 0.87   | 0.81  |
| DT         | 1.34      | 0.71  | 1.21   | 0.73  |
| BPNN       | 1.09      | 0.75  | 1.19   | 0.69  |
| SHAP–CNN   | 0.48      | 0.97  | 0.43   | 0.98  |



**Fig. 10** Comparison of training time and prediction time of four models



**Fig. 11** Comparison of surface temperature and heat island intensity of four models

model. SHAP can provide this interpretability, helping researchers and decision makers better understand and trust the model. At the same time, the urban heat island effect is influenced by multiple factors, and there may be complex interactions between these factors. SHAP can consider all possible feature combinations, accurately evaluate the contribution of each feature, and reveal the impact of these interactions on model predictions. Overall, the SHAP–CNN model provided accurate and reliable predictions for both surface temperature and heat island intensity, making it applicable for urban heat island monitoring and forecasting.

Through the high-precision urban heat island distribution map and feature contribution analysis output by the model, urban planners can accurately identify high-risk areas of the heat island, which often overlap spatially with socially vulnerable areas, such as low-income communities and old urban areas. Model analysis shows that in the old urban area of Xiamen Island, the contribution of building density to the heat island effect is 23% higher than that in the new area outside the island, while the regulating effect of vegetation cover is 18% lower. This finding provides a scientific basis for implementing differentiated intervention measures. At the policy application level, the model supports three key decision-making scenarios. First, through spatial overlay analysis of heat island risk and socio-economic data, priority can be given to the layout of cooling infrastructure in vulnerable communities; Second, based on the quantitative evaluation of feature contribution, targeted building density control and green space configuration standards can be formulated; Finally, the temporal evolution function of the prediction model can simulate the heat island mitigation effect under different policy scenarios, supporting scientific decision-making. The comparative study of

the model in Shanghai and Xiamen found that the same intensity of heat island effect can lead to higher health risks in densely populated old urban areas, indicating that policy formulation needs to take into account geographical fairness. Future research will further integrate socio-economic data and develop a heat island risk assessment module with a social equity perspective, providing technical solutions for environmental justice issues in the construction of "smart cities". These applications will contribute to achieving the United Nations Sustainable Development Goal (SDG11) of building inclusive, safe, and disaster resilient cities.

## Conclusion

The urban heat island effect has grown more severe in recent years, leading to a decline in residents' quality of life. However, traditional urban heat island monitoring techniques have struggled to balance efficiency and accuracy, making them unsuitable for practical deployment. To address this issue, this study put forward a hybrid model based on SHAP–CNN. This model used CNN to automatically extract key factors of the urban heat island effect and applied visualization tools to display the distribution and formation of heat islands. Experimental results showed that the F1 score of the proposed model increased from 0.38 to 0.93 after 500 iterations. The accuracy reached 88.7%, which outperformed the 62.1% accuracy of U-Net. After 600 iterations, the  $R^2$  value stabilized at 0.91. The model's RMSE eventually stabilized at  $0.43^\circ\text{C}$ . On the self-constructed data set of Shanghai, the RMSE reached  $0.48^\circ\text{C}$  and the  $R^2$  value was 0.97. The average training time was 2.49 h. The prediction results of land surface temperature and urban heat island intensity showed a high degree of agreement with actual values, with matching rates of 98.1% and 99.5%. These findings



validated that the proposed model is capable of efficiently and accurately monitoring and predicting the urban heat island effect while also assessing the heat island intensity across various urban regions. Especially, it points out that vulnerable communities often face overlapping risk factors, such as high building density and insufficient green space. These findings support prioritizing the allocation of limited cooling resources to environmentally justice sensitive areas, such as implementing targeted interventions such as rooftop greening and increasing shading facilities in old communities at high risk of urban heat islands. At the same time, the model prediction function allows planners to pre evaluate the improvement effect of different policy options on the community's thermal environment, avoiding the new unfairness brought about by the "one size fits all" governance. The promotion and application of this technology is expected to promote the transformation of urban climate adaptation strategies from "homogenization" to "precision" and "fairness", providing a new paradigm for urban sustainable development in the context of climate change.

However, there are still certain limitations in the research. First, the computational complexity of the model is relatively high, mainly due to the combined computational requirements of CNN's multi-layer convolution structure and SHAP algorithm; The second issue is that the training of the model relies on high-quality multi-source data, but some cities' historical remote sensing data have problems with low temporal resolution or incomplete spatial coverage, which may affect the model's migration performance in new areas. Third, the dynamic interaction process of urban microclimate, such as the real-time feedback mechanism between buildings, vegetation, and atmosphere, has not been fully considered, which to some extent limits the accuracy of ultrashort-term predictions. Therefore, in future research on model lightweighting, knowledge distillation techniques can be explored by training a compact student network guided by SHAP–CNN; We plan to develop a cross city transfer learning framework, utilizing data from Shanghai and Xiamen to pre train the basic model, and fine tune it with a small amount of labeled data from the target city. Finally, graph neural networks are introduced to model the real-time interactions of urban spatial elements.

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#### Declarations

#### Competing interests

The authors declare no competing interests.

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